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Evidence-backed exploratory analytics, KPI summary, and strategic findings

DATASET	RECORDS / SCOPE	REPORTING PERIOD	GENERATED DATE
sample_sales.csv	10 rows (8 cols)	2024-01-10 to 2024-01-30	18/10/2026

Executive Summary

Column 'quantity' contains 1 outlier values (10% of records) falling outside the IQR threshold interval [-0.5, 3.5]. Minimum is 1, maximum is 4, and median is 1.5.

- Outlier Detection in 'quantity': Column 'quantity' contains 1 outlier values (10% of records) falling outside the IQR threshold interval [-0.5, 3.5]. Minimum is 1, maximum is 4, and median is 1.5.
- Outlier Detection in 'unit_price': Column 'unit_price' contains 1 outlier values (11.11% of records) falling outside the IQR threshold interval [-600.235, 1160.125]. Minimum is 25, maximum is 1250, and median is 120.
- Temporal Trend: INCREASING (+150.01% over period): Activity tracked along 'transaction_date' spans 20 days (2024-01-10 to 2024-01-30) across 10 daily periods. The net period-over-period trajectory changed by 150.01% with a 'increasing' pattern. Unusual volatility was observed in periods: [2024-01-30].
- Concentration in 'product_category': 'Electronics' Dominance: The top category 'Electronics' in column 'product_category' represents 40% of all valid records (4 of 10 non-null rows). There are 4 distinct categories in total.
- Correlation Identified: 'unit_price' vs 'quantity' (r = -0.58): A statistically moderate negative linear association (Pearson r = -0.58, sample size n = 9) exists between 'unit_price' and 'quantity'.

Data Trust Rating: HIGH — Evaluated over 10 rows. Data completeness score is 98.75%. 2 quality notices identified.

Executive Key Performance Indicators

TOTAL RECORDS

10

Total validated rows analyzed across the dataset.

COMPLETENESS SCORE

98.8%

Ratio of non-null cells to total cells across all rows and columns.

MEAN UNIT_PRICE

$346.51

Arithmetic mean computed across 9 non-null values.

MEAN QUANTITY

1.80

Arithmetic mean computed across 10 non-null values.

PERIOD GROWTH RATE

+150.0%

+150.0% vs 2024-01-10 to 2024-01-30

Observed trend direction: increasing across 10 temporal buckets.

Data Quality & Completeness Scorecard

Overall Score: **98.8%** Status: **GOOD** Columns with Nulls: **1 / 8**

Column	Data Type	Completeness %	Null Count	Identified Issues	Status
unit_price	NUMERIC	90.0%	1	Column 'unit_price' exhibits 1 outliers (11.11% of values outside [-600.235, 1160.125]).	WARNING
quantity	NUMERIC	100.0%	0	Column 'quantity' exhibits 1 outliers (10% of values outside [-0.5, 3.5]).	INFO
transaction_id	CATEGORICAL	100.0%	0	None detected	INFO
customer_id	CATEGORICAL	100.0%	0	None detected	INFO
product_category	CATEGORICAL	100.0%	0	None detected	INFO
discount_applied	CATEGORICAL	100.0%	0	None detected	INFO
payment_status	CATEGORICAL	100.0%	0	None detected	INFO

Scorecard covers 100% of rows analyzed across all attributes. Null percentages reflect unpopulated cells.

Ranked Strategic Findings & Recommendations

#1 — Outlier Detection in 'quantity'

HIGH PRIORITY

OBSERVED FACT (EVIDENCE-BACKED)

Column 'quantity' contains 1 outlier values (10% of records) falling outside the IQR threshold interval [-0.5, 3.5]. Minimum is 1, maximum is 4, and median is 1.5.

EXPLORATORY HYPOTHESIS

Extreme values in 'quantity' could reflect high-value premium transactions, bulk institutional orders, or manual data entry anomalies.

RECOMMENDED ACTION

Conduct segmentation analysis excluding these 1 extreme records to compare baseline core behavior against premium outliers.

Limitations / Caveats: IQR boundaries assume unimodal distributions; highly multi-modal data may require domain-specific thresholds.

#2 — Outlier Detection in 'unit_price'

HIGH PRIORITY

OBSERVED FACT (EVIDENCE-BACKED)

Column 'unit_price' contains 1 outlier values (11.11% of records) falling outside the IQR threshold interval [-600.235, 1160.125]. Minimum is 25, maximum is 1250, and median is 120.

EXPLORATORY HYPOTHESIS

Extreme values in 'unit_price' could reflect high-value premium transactions, bulk institutional orders, or manual data entry anomalies.

RECOMMENDED ACTION

Conduct segmentation analysis excluding these 1 extreme records to compare baseline core behavior against premium outliers.

Limitations / Caveats: IQR boundaries assume unimodal distributions; highly multi-modal data may require domain-specific thresholds.

#3 — Temporal Trend: INCREASING (+150.01% over period)

HIGH PRIORITY

OBSERVED FACT (EVIDENCE-BACKED)

Activity tracked along 'transaction_date' spans 20 days (2024-01-10 to 2024-01-30) across 10 daily periods. The net period-over-period trajectory changed by 150.01% with a 'increasing' pattern. Unusual volatility was observed in periods: [2024-01-30].

EXPLORATORY HYPOTHESIS

Positive growth over this window may correspond to seasonality, marketing acquisition surges, or expanding customer activity.

RECOMMENDED ACTION

Forecast near-term capacity and supply chain requirements assuming continuation of current trajectory, validating against prior seasonal cycles.

Limitations / Caveats: Evaluates aggregate volume change between start and end intervals; external seasonal factors require multi-year historical data to establish cyclicality.

#4 — Concentration in 'product_category': 'Electronics' Dominance

MEDIUM PRIORITY

OBSERVED FACT (EVIDENCE-BACKED)

The top category 'Electronics' in column 'product_category' represents 40% of all valid records (4 of 10 non-null rows). There are 4 distinct categories in total.

EXPLORATORY HYPOTHESIS

Strong preference or volume concentration in 'Electronics' suggests either product-market fit or an over-reliance on a single segment.

RECOMMENDED ACTION

Evaluate operational dependencies on 'Electronics'. Determine if secondary segments ('Apparel', 'Home & Garden') can be expanded through targeted campaigns.

Limitations / Caveats: Category distribution reflects transaction frequency, which may differ from dollar-weighted revenue volume.

#5 — Correlation Identified: 'unit_price' vs 'quantity' (r = -0.58)

MEDIUM PRIORITY

OBSERVED FACT (EVIDENCE-BACKED)

A statistically moderate negative linear association (Pearson r = -0.58, sample size n = 9) exists between 'unit_price' and 'quantity'.

EXPLORATORY HYPOTHESIS

Higher values in 'unit_price' coincide with lower values in 'quantity', potentially indicating an inverse trade-off or discounting effect.

RECOMMENDED ACTION

Analyze paired scatter plots and residual distributions. Prior to making operational policy changes, execute a controlled A/B test or multivariate regression to verify whether an actual causal mechanism exists.

Limitations / Caveats: CRITICAL: Correlation does NOT demonstrate or imply causation. Unmeasured confounders or external factors may account for the relationship. Linear correlation assesses only linear co-movement and does not capture non-linear or threshold dependencies.

#6 — Operational Frequency: 'discount_applied' Baseline (false)

LOW PRIORITY

OBSERVED FACT (EVIDENCE-BACKED)

Operational baseline in 'discount_applied': 'false' represents 60% of valid records (6 of 10 non-null rows).

EXPLORATORY HYPOTHESIS

Standard operational workflow baseline where the predominant outcome is 'false'.

RECOMMENDED ACTION

Monitor 'discount_applied' for shifts in status rates, but treat this as a descriptive baseline rather than an executive revenue driver unless correlated with revenue loss or operational bottlenecks.

Limitations / Caveats: Status frequencies reflect operational states and are not primary drivers of commercial strategy.

#7 — Operational Frequency: 'payment_status' Baseline (Completed)

LOW PRIORITY

OBSERVED FACT (EVIDENCE-BACKED)

Operational baseline in 'payment_status': 'Completed' represents 80% of valid records (8 of 10 non-null rows).

EXPLORATORY HYPOTHESIS

Standard operational workflow baseline where the predominant outcome is 'Completed'.

RECOMMENDED ACTION

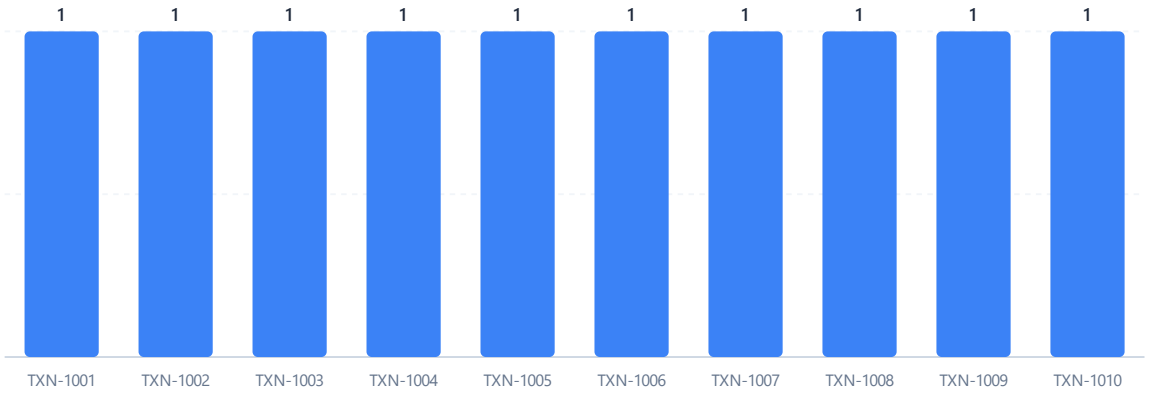
Monitor 'payment_status' for shifts in status rates, but treat this as a descriptive baseline rather than an executive revenue driver unless correlated with revenue loss or operational bottlenecks.

Limitations / Caveats: Status frequencies reflect operational states and are not primary drivers of commercial strategy.

Analytical Visualizations

Distribution by transaction_id

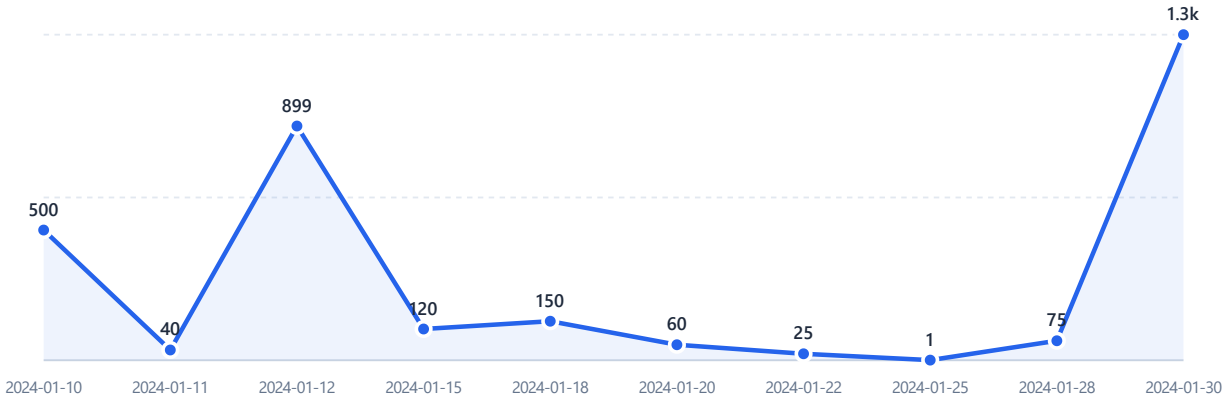
Frequency distribution across top 10 distinct categories



Shows volume concentration for column 'transaction_id'. Mode is 'TXN-1001' representing 10% of records. Note: counts represent volume, not revenue unless explicitly aggregated.

Temporal Trend: transaction_date

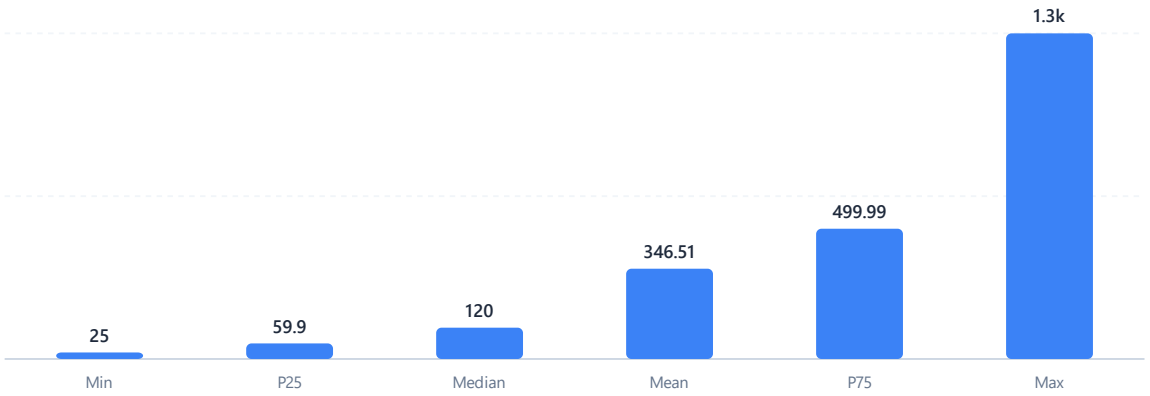
DAILY aggregation from 2024-01-10 to 2024-01-30



Temporal trend across 10 periods. Overall observed direction is 'increasing' with calculated growth rate of 150.01%.

Distribution Profile: unit_price

Five-number summary and mean computed via DuckDB



Distribution profile for 'unit_price'. Median = $120, Mean = $346.51, IQR = 440.09. 1 outliers identified (11.1%).

Detailed Tabular Summaries

Numeric Attributes Summary Statistics

Column	Count	Null %	Mean	Median	Std Dev	Min	Max	Outliers
unit_price	9	10.0%	$346.51	$120	445.59	$25	$1,250	1 (11.1%)
quantity	10	0.0%	1.80	1.50	1.03	1	4	1 (10.0%)

Full-dataset summary metrics computed via DuckDB analytical SQL engine. Outliers computed using Tukey's 1.5 IQR threshold.

Categorical Attributes Summary

Column	Distinct Values	Null %	Mode (Top Value)	Mode Count	Top Value Share
transaction_id	10	0.0%	TXN-1001	1	10.0%
customer_id	9	0.0%	CUST-201	2	20.0%
product_category	4	0.0%	Electronics	4	40.0%
discount_applied	2	0.0%	false	6	60.0%
payment_status	3	0.0%	Completed	8	80.0%

Categorical distribution summary. Unique ratios and mode concentrations computed across entire authorized dataset.

Statistical Distribution & Association Analysis

CRITICAL STATISTICAL NOTICE: Correlation measures mathematical association between variables in this dataset. It does not establish cause-and-effect relationships or directionality. Business decisions should not assume that changing one variable will produce changes in correlated variables without controlled experimentation or causal identification.

Temporal Trend Assessment: Time-series assessment over 'transaction_date' from 2024-01-10 to 2024-01-30 demonstrated a increasing direction across 10 reporting intervals.

Pairwise Pearson Correlation Analysis

Variable A	Variable B	Pearson r	Sample Size (N)	Statistical Interpretation
unit_price	**quantity**	**-0.584**	9	Observed Pearson correlation coefficient r = -0.584 indicates a moderate negative relationship across 9 paired observations.

Identified Statistical Outliers (Tukey IQR Criterion)

Column	Outlier Count	% of Records	Acceptable Bounds [Lower, Upper]	Business Implication
unit_price	1	11.1%	[-600.24, 1160.13]	1 data point(s) exceed statistical bounds [-600.24, 1160.13]. Review for recording anomalies or high-impact tail events.
quantity	1	10.0%	[-0.50, 3.50]	1 data point(s) exceed statistical bounds [-0.50, 3.50]. Review for recording anomalies or high-impact tail events.

Methodology, Limitations & Provenance

Analytical Engine: DuckDB Embedded SQL & Statistical Profiler

Data Profiling Method: Exhaustive streaming scan and exact SQL aggregation over complete dataset rows

Source File: sample_sales.csv

Analysis Execution Time: 179 ms

Statistical Appendix & Metric Definitions

Interquartile Range (IQR) Outlier Rule

`Lower Bound = Q1 - (1.5 × IQR); Upper Bound = Q3 + (1.5 × IQR)`

Identifies values that fall beyond 1.5 times the spread of the middle 50% of the distribution. Established by John Tukey for robust outlier detection resistant to extreme values.

Pearson Correlation Coefficient (r)

$r = \Sigma((x - \bar{x})(y - \bar{y})) / (\sqrt{\Sigma(x - \bar{x})^2} \times \sqrt{\Sigma(y - \bar{y})^2})$

Measures linear relationship strength between two continuous variables, bounded between -1.0 (perfect negative) and +1.0 (perfect positive). Does NOT imply causal relationship.

Completeness Score (%)

`(1 - (Total Null Cells / Total Cell Count)) × 100%`

Measures overall dataset fill rate across all recorded rows and columns. Scores above 95% indicate high data completeness.

Categorical Mode Concentration (%)

`(Frequency of Most Frequent Value / Total Non-Null Count) × 100%`

Indicates the dominance of the primary category. High concentration (>60%) flags potential class imbalance or monopolistic segments.